TECHRHYTHM 2.0

SyntaXsurvival Hackathon

Official Problem Statements

Phase 1: Online Round | April 2026

Organized by Code Craft Society & IDC, ITM University Gwalior

25	7	SyntaXsurvival 2026
Total Problem Statements	Domains Covered	Event

Instructions for Participants

1. Choose exactly ONE problem statement from any domain.
2. Prepare a 5–6 slide pitch deck using the official PPT template (available on Unstop).
3. Submit your PS number, team details, and pitch deck via the Google Form linked on the Unstop event page.
4. Late submissions will NOT be accepted. Shortlisted teams proceed to the 36-hour offline hackathon on April 08–10, 2026.

Domain Index

Domain	PS IDs	Count
Artificial Intelligence & Machine Learning	PS-01, PS-02, PS-03, PS-04	4
Cybersecurity	PS-05, PS-06, PS-07, PS-08	4
Sustainability & Environment	PS-09, PS-10, PS-11, PS-12, PS-13	5
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Artificial Intelligence & Machine Learning

4 Problem Statements

PS-01 Smart Resume Screening with Bias Detection

Background	HR teams spend 80% of their time screening resumes manually, often with unconscious bias based on gender, college name, or background. This leads to talent being overlooked and opportunities being distributed unfairly.
Problem	Develop an AI-based resume screener that ranks candidates fairly, detects potential bias, and explains why a candidate was selected or rejected.
Objective	To make the hiring process faster, fairer, and more transparent using AI-driven candidate evaluation.

PS-02 AI-Powered Sign Language to Speech Translator

Background	In India, millions of people struggle with hearing and speech impairments, but most people do not understand sign language, creating a severe communication gap in everyday interactions.
Problem	Develop an ML model that can recognize Indian Sign Language (ISL) in real-time using a camera and convert it into text or audio output.
Objective	To eliminate the communication barrier between sign language users and the general public.

PS-03 Agentic AI for Rural Healthcare Triage

Background	In many rural areas, the doctor-to-patient ratio is critically low. Patients often travel hours for conditions that could be managed locally, or worse — ignore life-threatening symptoms entirely due to lack of access.
Problem	Build an NLP-based AI triage system that operates in local languages, supports low-connectivity environments, and assists rural health workers in identifying patients who need urgent specialist care.
Objective	To bridge the healthcare access gap in rural India by enabling intelligent, language-inclusive, and reliable first-level medical triage.

PS-04 Water Bodies Monitoring Using AI

Background	Water bodies such as rivers, lakes, and reservoirs are facing increasing pollution, declining water quality, and ecological imbalance. Traditional monitoring methods are slow, manual, and not real-time.
Problem	There is no efficient real-time system to continuously monitor water quality parameters such as turbidity, pH, temperature, and contamination levels. Build an AI-powered system that bridges this gap using sensor data and intelligent analytics.

Objective	To build an AI-powered monitoring system that collects real-time data from sensors, analyzes water quality, and provides automated alerts for pollution or abnormal conditions.

Cybersecurity

4 Problem Statements

PS-05 Phishing URL and Email Detection Extension

Background	Phishing is the most common cybercrime method in India, with over 5,000 phishing websites reported monthly. Students and senior citizens are the most vulnerable, often clicking malicious links embedded in realistic-looking emails or messages.
Problem	Build a browser extension and email scanner that uses ML to analyze URLs, email headers, and content patterns in real-time to detect phishing attempts and warn users before they interact with them.
Objective	To protect everyday internet users from phishing attacks through intelligent, real-time threat detection integrated directly into their browsing and email experience.

PS-06 Intelligent Intrusion Detection System

Background	With increasing digital transformation, cyber threats such as hacking, malware, and unauthorized access are growing rapidly. Conventional intrusion detection systems rely on predefined rules and signatures, making them ineffective against new or evolving cyberattacks.
Problem	Create an AI-based intrusion detection system that can analyze network traffic, identify anomalies, and detect both known and unknown cyber threats in real-time.
Objective	To develop a smart, adaptive IDS that goes beyond signature-based detection and actively learns from network behavior to counter emerging threats.

PS-07 Security Challenges in IoT Networks

Background	With the rapid growth of IoT, billions of devices — smart home systems, wearables, and industrial sensors — are connected to the internet. Most have limited computational power and weak security mechanisms, making them highly vulnerable to cyberattacks.
Problem	Design a secure and lightweight system that can detect, prevent, and respond to cyber threats in IoT networks — including unauthorized access, device spoofing, and DDoS attacks — while ensuring data privacy and efficient performance on resource-constrained devices.
Objective	To develop a scalable, lightweight cybersecurity solution purpose-built for the unique constraints of IoT infrastructure.

PS-08	Deepfake Video & Audio Detector
Background	The use of deepfake technology is increasing rapidly, leading to fake news, identity theft, and manipulation of public opinion. Digital content can no longer be trusted at face value.
Problem	Develop an ML tool that analyzes video and audio patterns to detect AI-generated (deepfake) content with high accuracy and provides an authenticity confidence score.
Objective	To verify the authenticity of digital media content and protect individuals and institutions from deepfake-based misinformation.

Sustainability & Environment

5 Problem Statements

PS-09	AI-Based Waste Segregation Assistant
Background	India generates over 62 million tonnes of municipal solid waste annually, but less than 20% is properly segregated at source. Citizens lack awareness and real-time guidance on how to classify and dispose of waste correctly.
Problem	Build a mobile app where users can photograph waste items and receive instant AI-based classification into dry, wet, hazardous, or e-waste categories, along with nearby disposal point mapping.
Objective	To promote responsible waste disposal habits among citizens through accessible, AI-driven waste classification and real-time disposal guidance.

PS-10	Integrated E-Waste Circular Economy Marketplace
Background	Most e-waste (phones, laptops) ends up in landfills because there is no formal link between the individual consumer, the scrap collector, and the certified recycler. Residual material value is lost entirely.
Problem	Create a unified marketplace that incentivizes citizens to dispose of e-waste responsibly. The platform should calculate the residual value of a device based on its components (gold, copper, lithium) and connect users with certified recyclers.
Objective	To formalize and incentivize e-waste disposal by creating a transparent, value-driven marketplace that benefits consumers, recyclers, and the environment.

PS-11	AI-Driven Smart Grid Load Balancer for Rural Electrification
Background	In rural India, power grids often face instability due to unpredictable load shedding and the integration of decentralized solar grids. Unbalanced loads frequently lead to transformer failures and inconsistent supply.
Problem	Develop an intelligent load-balancing system that predicts peak usage hours in village clusters and automatically suggests power rerouting or non-essential load shedding (e.g., streetlights, heavy machinery) to prevent grid collapse.

Objective	To improve the reliability and efficiency of rural power supply using ML-based demand forecasting and real-time grid stability monitoring.

PS-12	**Campus Carbon Footprint Tracker**
Background	Universities are significant energy consumers through air conditioners, servers, lighting, and transport — all contributing to carbon emissions. Most institutions have no quantified understanding of their actual carbon footprint.
Problem	Create a web platform that collects energy usage data from campus departments, calculates total carbon emissions, benchmarks against national standards, and recommends actionable reduction strategies.
Objective	To help educational institutions measure, understand, and actively reduce their environmental impact through data-driven carbon management.

PS-13	**Sustainable Fertilizer Usage Optimizer for Higher Yield**
Background	Excessive and improper use of fertilizers leads to soil degradation and reduced agricultural productivity, negatively impacting farmers' income and long-term land health.
Problem	Create a data-driven solution that recommends optimal fertilizer types and quantities based on soil health, crop type, and weather patterns — ensuring sustainable agricultural practices.
Objective	To reduce soil degradation and improve crop yield by providing farmers with personalized, science-backed fertilizer recommendations.

HealthTech & MedTech

5 Problem Statements

PS-14	**Mental Health Chatbot for College Students**
Background	One in five college students in India suffers from anxiety or depression, yet most never seek help due to stigma, cost, or lack of access. University counseling centers are understaffed with long waiting times.
Problem	Build a 24/7 empathetic AI chatbot that provides mental wellness support, mood tracking, and breathing exercises, and escalates to a human counselor when it detects a crisis situation.
Objective	To make mental health support accessible, stigma-free, and available around the clock for college students across India.

PS-15	**Tele-ICU Monitoring System for Tier-3 Hospitals**

Background	Small-town hospitals have ICU beds but lack 24/7 specialist doctors. Critically ill patients often die during transit to metro cities — a problem of geography and resource scarcity.
Problem	Develop a Tele-ICU interface where a single specialist in a central hub can monitor real-time vitals (ECG, SpO2, Heart Rate) of up to 50 patients across different remote hospitals simultaneously.
Objective	To save lives in resource-constrained hospitals by enabling remote specialist oversight through a high-concurrency, real-time patient monitoring platform.

PS-16	Medicine Expiry and Inventory Tracker for Rural Clinics
Background	Rural Primary Health Centers frequently administer expired medicines or face stock-outs due to poor inventory management. There is no digital system to track stock levels and expiry dates in most centers.
Problem	Develop a simple offline-first web or mobile app where clinic staff can scan or enter medicine inventory, receive expiry alerts, auto-generate reorder requests, and share reports with district health officials.
Objective	To improve medicine management in rural healthcare facilities, preventing expired medicine use and ensuring consistent drug availability.

PS-17	Blood Donation Matching & Emergency Alert System
Background	During medical emergencies, finding the right blood group takes too long, which can be life-threatening. Existing systems are fragmented, slow, and not real-time.
Problem	Create a real-time dashboard that connects patients with nearby available blood donors and sends instant push notifications, with location-aware matching and availability status.
Objective	To reduce the time required for blood procurement in emergencies and save lives through intelligent, location-based donor-patient matching.

PS-18	Privacy Preservation in Healthcare Using Federated Learning
Background	Healthcare data is sensitive and subject to strict regulations (HIPAA, GDPR). Traditional ML methods require data to be compiled centrally, increasing privacy risks. Multi-institutional collaboration through data pooling is essential for robust AI models, but is often not feasible due to patient privacy concerns and institutional data-sharing policies.
Problem	Build a Federated Learning-based Healthcare Privacy Preservation System that allows multiple hospitals to collaboratively train AI models without ever sharing raw patient data — only model weights or gradients are exchanged with a central aggregation server.
Objective	To enable secure, privacy-preserving multi-institutional AI model training in healthcare, ensuring that sensitive patient records (EHRs, medical images, diagnostic reports) never leave the premises of the originating institution.

Smart Education

3 Problem Statements

PS-19	Real-Time Student Drop-Out Prediction & Intervention System
Background	In rural schools, students often drop out due to economic or social reasons. By the time the administration notices, it is too late to bring the student back and the opportunity for early intervention is lost.
Problem	Build a predictive analytics tool for school boards that analyzes attendance patterns, mid-term grades, and social indicators to flag students at high risk of dropping out.
Objective	To enable proactive, early-stage intervention by teachers and administrators, reducing drop-out rates through data-driven early warning signals.

PS-20	AI-Powered Career Counselling Platform for Schools
Background	A lack of adequate career counselling in schools contributes to poor career choices among students, leading to mismatched skills, job dissatisfaction, and unemployment. Many students make decisions based on limited information or societal pressure.
Problem	Develop an AI-powered platform that provides personalized career guidance based on students' interests, strengths, and market trends — accessible to schools in Tier-2 and Tier-3 cities.
Objective	To help students make informed, data-backed career choices that align with their abilities and the demands of the job market, in line with NEP 2020.

PS-21	Unified College ERP Dashboard for Students
Background	Students across most Indian universities juggle five to six different portals for attendance, results, fees, hostel, and notices — none of which are interconnected, causing confusion, missed deadlines, and wasted time.
Problem	Build a unified, responsive web dashboard that aggregates all college services into one smart interface with notifications, deadline tracking, and a personal academic calendar.
Objective	To improve the student experience by centralizing all academic and administrative services into a single, easy-to-use platform.

FinTech

2 Problem Statements

PS-22	Micro-Loan Credit Scoring for the Unbanked
Background	Over 190 million Indians remain unbanked or underbanked with no credit history, making them ineligible for formal loans. They frequently fall prey to informal moneylenders with exploitative interest rates.
Problem	Develop an alternative credit scoring system using non-traditional data such as mobile usage, utility payments, and UPI transactions to assess loan eligibility and connect users with micro-lending platforms.
Objective	To create a fair and inclusive credit evaluation system that extends formal financial services to underserved populations.

PS-23	Real-Time UPI Fraud Detection System
Background	UPI transactions in India crossed 13 billion monthly in 2024, but fraud cases are rising rapidly. Most fraud detection systems respond after the transaction is complete, causing irreversible financial loss.
Problem	Build a real-time transaction monitoring system that uses ML to detect anomalous UPI behavior and flags or halts suspicious transactions before they are processed.
Objective	To protect users from financial fraud by intercepting suspicious UPI transactions in real-time using intelligent pattern recognition.

IoT & Hardware

2 Problem Statements

PS-24	Smart Helmet for Construction Workers
Background	Construction sites report thousands of accidents annually in India, often due to workers removing helmets in heat or fatigue. There is no real-time monitoring of safety compliance on most sites.
Problem	Design an IoT-enabled smart helmet that detects if it is being worn, monitors head impact, body temperature, and worker fatigue, and sends real-time alerts to a site supervisor's dashboard.
Objective	To reduce construction site fatalities and injuries through continuous, automated worker safety monitoring.

PS-25	Automated Smart Irrigation System for Small Farms
Background	India is one of the largest consumers of groundwater globally, with 80% consumed by agriculture — much of it wasted through over-irrigation due to guesswork. Small farmers lack affordable smart solutions.
Problem	Build a low-cost IoT-based irrigation system using soil moisture sensors, weather APIs, and automated water valves that irrigates crops only when needed and sends control alerts via SMS or app.
Objective	To conserve water resources and improve agricultural efficiency for small and marginal farmers using affordable IoT technology.

For queries: techfest@itmuniversity.ac.in | Register & Submit on: Unstop

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